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HY Pathology, by Dr Michael D Mehlman

The focus of this PDF is a HY review of **general pathology** principles for USMLE. When we talk about “Path,” every organ system has countless diseases and mechanisms that could be expounded upon. Recognize that my other subject PDFs (i.e., HY Cardio, Pulm, Renal, Gastro, etc.) are all “Path,” where I go into extensive detail about the things you need to know for USMLE. I reiterate / preface this way because if you’re currently looking for in depth Path in a specific subject area, use my other subject PDFs. Going through all of my PDFs will cover all of your “Path” for USMLE. This PDF is just general pathology principles once again.

HY General Pathology

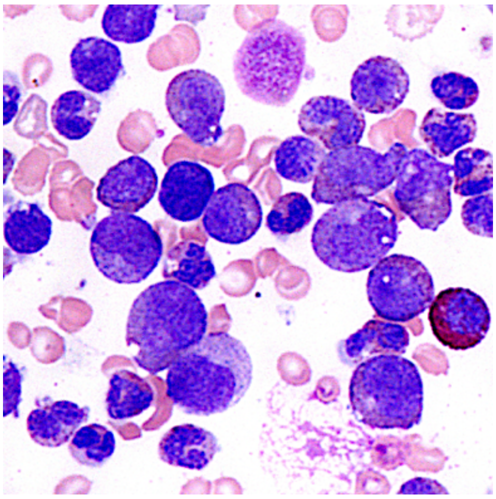
HY Path terms	
Anaplasia	- Reversion to more immature cell type (i.e., a cell goes from, e.g., mature skin cell back to a stem cell). - Means high-grade in cancer and poorer prognosis.
Aplasia	- Tissue/organ doesn't develop. - DiGeorge syndrome → aplasia of 3rd and 4th pharyngeal pouches → missing thymus and parathyroid glands → T-cell deficiency and hypocalcemia/hyperphosphatemia.
Atrophy	- Diminished cell growth/size due to under-stimulation. - Shows up on NBME exam for effect on prostate tissue treated with orchiectomy in the setting of prostatic adenocarcinoma; ↓ testosterone → ↓ DHT → ↓ prostate stimulation. - Testicular atrophy in the setting of exogenous anabolic steroid use; ↑ negative feedback at hypothalamus/anterior pituitary → ↓ LH secretion → ↓ testicular stimulation. - Thyroid follicular atrophy with exogenous T3/T4 use; ↑ negative feedback at hypothalamus/anterior pituitary → ↓ TSH secretion → ↓ thyroid stimulation. - Menopausal effects on ovaries and endometrium; don't confuse with menses, which are apoptotic.
Dysplasia	- Cellular atypia that is precursor to cancer; cells are not yet cancerous/tumorigenic. - Reversible process; students often erroneously think it is irreversible. - USMLE is not going to ask, "Is dysplasia irreversible or reversible?" What they will do is give you, e.g., koilocytes with HPV 16/18 infection, and you need to know most cases of low- and high-grade squamous epithelial dysplasia are reversible / spontaneously regress.
Hyperplasia	- Increased cell number. - Endometrial hyperplasia due to unopposed estrogen is highest yield example on USMLE. Anovulatory cycles → no corpus luteum → no progesterone production → unopposed estrogen → endometrial hyperplasia → increased risk of endometrial dysplasia and adenocarcinoma. I discuss these mechanisms in high detail in my Obgyn / Repro PDF. - Benign prostatic hyperplasia (BPH) → older males will have large, hyperplastic prostates due to lifetime of DHT exposure.
Hypertrophy	- Increased cell size. - Highest yield example on USMLE is ventricular myocardial hypertrophy. High afterload causes concentric hypertrophy; high preload causes eccentric hypertrophy. I discuss these mechanisms in the Cardio section and HY Arrows PDF. - Skeletal muscle hypertrophy in response to resistance weight training.
Metaplasia	- One mature cell type becomes another mature cell type; reversible. - Barrett esophagus caused by reflux is highest yield example on USMLE: non-keratinized stratified squamous epithelium of distal esophagus → intestinal columnar epithelium (meaning has goblet cells that produce mucous). The stomach doesn't have goblet cells; it has mucous neck cells, aka foveolar cells.
Neoplasia	- Irreversible conversion of a cell to a tumorigenic one that grows uncontrollably. - This does not necessarily mean conversion of a cell to a cancerous one. The term cancer means malignant potential (i.e., capable of metastasis). But benign tumors such as fibroadenoma and uterine fibroids are still neoplastic. - Cervical intraepithelial neoplasia (CIN) is often reversible, as it is technically dysplasia, despite the name.

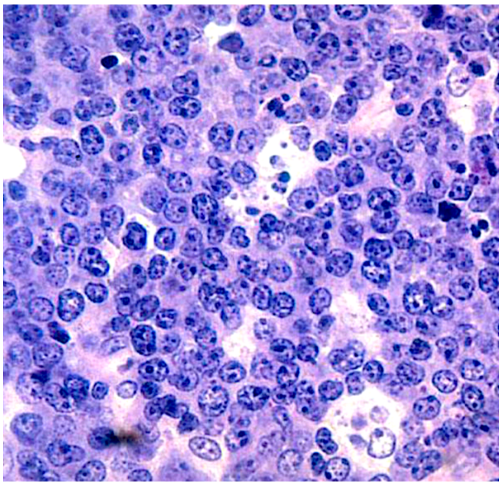
Apoptosis versus necrosis	
Apoptosis	- Cell death that is programmed and well-controlled; active process; uses ATP.
	- **Intrinsic pathway:** Activated in response to cell damage/stress/infection: Bax and Bak are intracellular proteins that create pores on the surface of mitochondria → causes leakage of cytochrome c from mitochondria to cytosol → cytochrome c triggers assembly of protein complex called apoptosome → **activates caspases** → caspases act as "molecular scissors," breaking down cellular structures → cell death.
	- **Extrinsic pathway:** Activated in response to external signals: death receptors (FAS or TNF receptor) are activated on the cell surface via external death ligands (FAS-L or TRAIL) → induces formation of death-inducing signaling complex (DISC) → **activates caspases.**
	- USMLE wants you to know hepatocellular death in hepatitis is due to T-cell-mediated apoptosis. Direct viral cytopathicity is wrong answer. This is also asked on the NBME where they show a histo pic of hepatocytes during hepatitis infection + they ask what's occurring here → answer = "apoptosis caused by activation of death receptor extrinsic pathway"; wrong answer is "apoptosis caused by activation of mitochondrial intrinsic pathway." - For whatever magical reason, you need to be aware that "DNA ladders of 180bp" can be formed during apoptosis. Not my opinion. On random offline NBME. - Apoptotic cells will fragment into smaller apoptotic bodies, which are phagocytosed. - Menstruation is apoptosis, not atrophy. Menopause is atrophy.
Necrosis	- Cell death that is uncontrolled, usually from injury or ischemia; passive process; does not use ATP. Necrosis types:
	- **Coagulative** → cellular architecture is maintained. - Myocardial infarction and acute tubular necrosis are HY examples.
	- **Liquefactive** → cellular architecture not maintained + dissolved by hydrolytic enzymes. - Refers to **1) abscesses and 2) anything CNS-related.**
	- Caseous → cheese-like necrosis. - Refers to 1) TB; 2) fungal infections; and 3) *Bartonella henselae* (cat-scratch disease).
	- **Enzymatic-fat** → Enzymatic fat necrosis = acute pancreatitis → pancreatic lipases dissolve surrounding architecture + chelate Ca^{2+} → saponification (soap formation).
	- **Non-enzymatic fat** → breast trauma.
	- **Fibrinoid** → "-Oid" means "looks like but ain't" → therefore looks like fibrin, even though it ain't fibrin; seen in polyarteritis nodosa, which affects small- and medium-sized arteries.
	- **Dry gangrenous** → tissue death (usually limbs) due to interrupted blood supply. - Technically a type of coagulative necrosis, but I've seen NBME list them as separate answers. - Classic example is diabetes, where neuropathy leads to patient not being able to feel his or her feet → foot injury → then vasculopathy leads to inability to heal the lesion → gangrene. - Can also refer to Buerger disease (thromboangiitis obliterans), which is digital gangrene in a heavy smoker.
	- **Wet gangrenous** → infective necrosis of limbs; can sometimes be a sequela of dry gangrene. - Fournier gangrene is necrosis of perineum/scrotum in advanced diabetes (on NBME).
	- **Gas gangrenous** → *C. perfringens* secretes α-toxin/phospholipase that produces CO_2 gas within tissues → crepitus/crunching of necrotic skin (subcutaneous emphysema). - Can also occur within the gall bladder (emphysematous cholecystitis) due to *C. perfringens.*
	- **Cystic medial necrosis** → necrosis of large arteries, where collagen-linking defects occur with the precipitation of cyst-like lesions within arterial wall. - Results in aortic dissection and aneurysm. - Seen in connective tissue disorders like Marfan and Ehlers-Danlos, and systemic hypertension. - NBME can write the answer as just "medial necrosis," sans the cystic, but it's the same thing.

Neuro necrosis points
- Liquefactive necrosis is the answer for necrosis of nervous system tissue. - "Red neurons" is the answer for what will be seen acutely with ischemic infarction of the CNS. This refers to their strong eosinophilic (pink) staining with H&E. This is in contrast to myocardial infarction, where "no histologic changes" is the answer immediately after. - Microglia are the resident macrophages of the CNS. They phagocytose necrotic brain/spinal tissue. - Astrocytes are the glial cells (non-neuronal cells of CNS/PNS) that proliferate and become a glial scar (gliosis). In other words, if they ask which cell is responsible for scar formation in CNS, answer = astrocyte.

Wallerian degeneration
- Wallerian degeneration is a term that refers to degradation of an axon/myelin sheath distal to the site of injury. Regrowth occurs at maximum of 1mm/day. Both PNS and CNS neurons undergo Wallerian degeneration, but regeneration occurs within the PNS, not CNS. This is because PNS Schwann cells can regenerate myelin, whereas CNS oligodendrocytes do not effectively regenerate myelin post-injury. - The optic nerve is considered an extension of the CNS and is myelinated by oligodendrocytes; the other cranial nerves are part of the PNS and are myelinated by Schwann cells. This distinction is why degeneration of the optic nerve results in permanent blindness, whereas other facial nerves (e.g., CN VII for Bell's palsy) have better regenerative potential. The caveat, however, is that the extent of CN recovery depends on the nature of the injury and is not always complete, the same way nerve injuries in the limbs might not fully recover either.

Cell injury points for USMLE	
Reversible	**Irreversible**
- Blunting of microvilli. - Cellular / mitochondrial swelling. - Cellular blebbing. - Decrease in intracellular ATP concentrations. - Disaggregation of ribosomes. - Disruption of cytoskeletal elements. - Fatty change (steatosis in the liver). - Glycogen accumulation. - Loosening of intercellular attachments. - Metastatic calcification (deposition of calcium in tissues in the setting of hypercalcemia). - It is in my view going through NBME exams that **cellular swelling** is highest yield for this point, particularly with regard to hypoxic injury of renal PCT cells – i.e., cells swell as damage occurs prior to overt tubular necrosis. This is because of reduced Na^+/K^+ ATPase activity that leads to Na^+ buildup within the cell → retention of water → swelling. NBME also asks for ↓ Na^+/K^+ ATPase activity as the answer for "hydropic changes" in renal PCT cells. - Cellular swelling can also occur with myocardium. NBME will give a patient who's died from MI and cellular swelling is observed on biopsy; they ask why → answer = ↓ Na^+/K^+ ATPase activity. In other words, although both cellular swelling and ↓ Na^+/K^+ ATPase activity are reversible, just be aware NBME still likes to ask this same mechanism posthumously in MI patients.	- Cellular autolysis (self-digestion of the cell by lysosomal enzymes). - Disruption of plasma membrane / increased membrane permeability / cell lysis. - Dystrophic calcification (deposition of calcium salts within necrotic tissues). - Karyolysis (dissolution of the nucleus). - Karyorrhexis (fragmentation of the nucleus into small, irregularly shaped fragments). - Loss of organelle integrity. - Pyknosis (nuclear condensation and shrinkage).

HY Cancer Genes for USMLE	
APC	- Mutations cause Familial Adenomatous Polyposis (FAP); chromosome 5; AD. - Hundreds to thousands of polyps on colonoscopy; 100% cancer risk. - FAP + soft tissue (e.g., lipoma) or bone tumors (e.g., of the skull) = Gardner syndrome. - FAP + CNS tumors = Turcot syndrome. - Answer for Tx on 2CK Peds form is total proctocolectomy in 18-year-old (presumably since 100% chance of cancer + too difficult to conservatively screen).
BCL-2	- Overexpressed in follicular lymphoma (most common indolent non-Hodgkin lymphoma) as a t(14;18) translocation. - Codes for anti-apoptotic molecule. - Follicular lymphoma will present as waxing/waning painless lateral neck mass over 1-2 years in an adult. - USMLE might give research-type Q where *BCL-2* is overexpressed (unrelated to follicular lymphoma) and they ask what would be expected → answer = increased lifespan of population of cells (makes sense, since anti-apoptotic molecule).
BCR-ABL	- HY for chronic myelogenous leukemia (CML). - t(9;22) translocation of these two genes (aka Philadelphia chromosome) codes for a "fusion protein." - This fusion protein = an oncogenic tyrosine kinase. - CML is answer for leukemia when the Q gives you lots of myelo-sounding cells (i.e., myelocytes, promyelocytes, metamyelocytes). 4/5 Qs on USMLE that mention these cells are CML. Leukocyte ALP will be low.  - The blood smear for CML I refer to as a "motley mix," or a "soup." This is a buzzy image on NBME for CML. - Imatinib is Tx for CML. It can cause fluid retention / peripheral edema. - The MOA of imatinib = targets *BCR-ABL* tyrosine kinase.
BRAF	- Proto-oncogene that can be seen in some melanomas. - Codes for serine-threonine kinase.
BRCA1/2	- Breast cancer tumor suppressor genes. - NBME exam asks for what the gene does → answer = "recombinational ds-DNA repair." - Can also lead to gynecologic cancers in females or breast/testicular cancers in males. - Answer on an NBME exam for appropriate prophylaxis in patient who has confirmed mutation is "bilateral oophorectomy and mastectomy."
c-KIT	- Proto-oncogene; can be mutated in some gastrointestinal stromal tumors (GIST).
c-MYC	- Overexpressed in Burkitt lymphoma (a type of NHL) as a t(8;14) translocation. - Codes for a transcription factor. - Burkitt lymphoma will be mass of the jaw or abdomen.

	- Histo is buzzy for "starry sky" appearance, which is a basophilic (purple) background of B cells with scattered translucent macrophages.  - The macrophages are referred to as "tingible body" macrophages (correct, not tangible), where **apoptosis** occurs. This is asked on an NBME exam, where they have an arrow that points to one of the macrophages, and the answer is apoptosis.
EGFR	- Non-small cell lung cancer (adenocarcinoma) and glioblastoma multiforme. - Codes for EGFR tyrosine kinase. - Erlotinib asked on USMLE → targets EGFR tyrosine kinase.
HER2/neu (ERBB2)	- Tyrosine kinase expressed in some breast cancers. If (+) → poor prognostic indicator. - Trastuzumab (Herceptin) is a drug that targets *HER2/neu*. It can cause cardiotoxicity.
JAK2	- Can be activated in polycythemia vera, essential thrombocytosis, and myelofibrosis. - USMLE wants increased proliferation of hematopoietic stem cells, not pluripotent stem cells, for PV. Patient will have high RBCs, PLUS either high WBCs and/or platelets (i.e., 2-3 of the cell lines will be high in Qs, but RBCs are always high). Generalized pruritis after showers can be seen from basophilia. Hyperviscosity syndrome (i.e., blurry vision, headache, Raynaud) from high RBCs is treated with phlebotomy. Hydroxyurea can decrease recurrence of episodes. EPO is *low*, not high, in PV. This is due to negative feedback, where high RBC production suppresses EPO. - Essential thrombocytosis will present in Qs as platelets over a million (NR 150-450,000). They will say there's pain or discoloration in tips of fingers or give hyperviscosity-type findings. Bone marrow will show increased megakaryocytic proliferation (this latter finding will not be seen in reactive thrombocytosis, which is high platelets from infection). - Myelofibrosis will present with teardrop-shaped RBCs (dacrocytes) and/or "dry tap" on bone marrow aspiration. Massive splenomegaly is seen in basically all questions. NBME can write the answer as simply "defective hematopoiesis" for myelofibrosis.
KRAS	- Proto-oncogene; codes for a GTPase. - The GTPase mutation means it is constitutively active (i.e., always active / cannot shut off). - The answer on USMLE for the first gene mutated in colonic neoplasia. Usually starts as a dysplastic polyp prior to progression to overt colon cancer. - Colon cancer often develops as a result of *progressive* mutations, rather than one mutation straight-up. In other words, first *KRAS*, then *PTEN*, then *DCC*, then *TP53*. - If they tell you a colon cancer has metastasized and force you to choose a gene that's mutated, go with *TP53* (codes for p53 protein). - If they tell you a polyp is seen and there is no evidence of invasion of the stalk, choose *KRAS*. This is on NBME exam.
MEN1	- Mutations cause Multiple Endocrine Neoplasia type I. - 3Ps → Pituitary tumor (e.g., prolactinoma), Parathyroid adenoma (or diffuse 4-gland hyperplasia), and Pancreatic tumor (e.g., gastrinoma; Zollinger-Ellison syndrome). - HY point is that NBME Qs need not give all findings within a MEN syndrome. For example, the Q can give gastrinoma alone and ask for the gene → *MEN1*.

MSH2/6 *MLH1* *PMS2*	- Hereditary non-polyposis colorectal cancer (HNPCC). - Mismatch repair genes; mutations cause "microsatellite instability." - Colonic polyps/cancer; also associated with gynecologic cancer.
NF1	- Neurofibromatosis type I; chromosome 17; AD. - Café au lait spots (hyperpigmented macules), axillary/groin freckling, neurofibromas (nerve sheath tumors presenting as nodules under the skin), weird CNS tumors (i.e., oligodendroglioma, ependymoma, meningioma), **optic glioma** (CN II tumor). - Classic disease that demonstrates **variable expressivity,** which means varying disease severity (I talk more about this stuff in my HY Genetics PDF).
NF2	- Neurofibromatosis type II; chromosome 22; AD. - Presents with acoustic schwannoma. Can be bilateral. - Sometimes meningiomas.
RB	- Congenital retinoblastoma (i.e., leukocoria in 1-year-old) and osteosarcoma. - Tumor-suppressor gene; autosomal dominant (two hits required, but chance of second mutation is 100%, so AD not AR). - RB protein is normally in a complex with E2F, a transcription factor, repressing it and preventing cell cycle progression. CDK/Cyclin complexes then phosphorylate RB, releasing it from E2F. E2F then goes to the nucleus and transcribes genes → cell cycle progression. - NBME simply wants you to know that decreased RB phosphorylation is therefore a **wrong** answer for what we'd expect in cancer cells.
RET	- Proto-oncogene → MEN 2A/2B. - Both MEN 2 syndromes have pheochromocytoma and medullary thyroid carcinoma. - MEN 2A has parathyroid adenoma or hyperplasia (same as MEN 1). - MEN 2B has Marfanoid body habitus and/or mucosal neuromas. - Same as with MEN 1, a HY point is that NBME Qs need not give all findings within a MEN syndrome. For instance, one NBME Q just gives medullary thyroid carcinoma alone, and the answer is *RET*. Another gives groin pain (urolithiasis due to hypercalcemia) + a neck tumor, and the answer is *RET*.
TP53	- Codes for p53 tumor-suppressor protein; halts the cell cycle in the setting of cellular damage so that DNA repair can occur. - Li-Fraumeni syndrome = congenital mutation in *TP53* leading to cancers of various organ systems. USMLE wants "failure of regulation of apoptosis" as the answer. "Defective DNA repair enzymes is wrong answer," since p53 itself does not directly repair DNA. - HY answer on NBME for the gene mutated in cancer that has metastasized. - Pleiotropy = one gene has multiple effects; *TP53* is textbook example, since mutation can cause many unrelated cancers, such as pancreatic, ovarian, colon, etc.
TSC1/2	- Tuberous sclerosis; autosomal dominant; hamartin and tuberin proteins. - Intellectual disability; periventricular nodules (tubers; aka cortical hamartomas; can present as seizures); adenoma sebaceum (aka angiofibromas, which are skin-colored/reddish papules on cheeks, nose, and in nasolabial folds); subungual fibromas (nailbed tumors); cardiac rhabdomyoma (ball-in-valve murmur that presents as diastolic rumble that attenuates with change of positioning); renal angiomyolipoma.
VHL	- Autosomal dominant; chromosome 3. - **Renal cell carcinoma, PLUS retinal and/or cerebellar hemangioblastomas.** - Pancreatic cysts can also be seen (on NBME). - The RCC need not be bilateral; don't confuse with angiomyolipoma of TSC. - New NBME Q shows picture of gross kidney lesion + tells you patient has cerebellar hemangioblastoma + retinal angioma; they ask what kind of kidney lesion they're showing → answer = "renal adenocarcinoma"; angioma is wrong answer.
WT1	- Can be isolated Wilms tumor; the answer on USMLE for kidney tumor in a kid almost always; presents as painless flank mass in 2-4-yr old. However Wilms tumor can also present as part of constellation syndromes: - Denys-Drash syndrome = gonadal agenesis + Wilms tumor. - WAGR syndrome = Wilms tumor, Aniridia, Genitourinary abnormalities, Retardation (can be associated with other genes as well, but just know that WT1 is associated).

	- Beckwith-Wiedemann syndrome = fetal macrosomia, macroglossia, hemihypertrophy, hypoglycemia, and Wilms tumor. Harder NBME Q gives big baby with hemihypertrophy + doesn't mention Wilms tumor; they ask what else could be seen → answer = hypoglycemia.

Other HY Genes for USMLE	
ATP7B	- Wilson disease; autosomal recessive; chromosome 13. - Copper overload. - Inability to secrete copper into bile from the liver. Copper is normally excreted by the body via secretion into bile. - ↑ urinary copper + ↓ serum ceruloplasmin (copper-binding protein in the blood; in the case of copper overload, body tries to minimize amount carried in blood). - Buzzy / pass-level detail is Keiser-Fleischer rings, which is copper deposited in the cornea of the eye. Vignette can give you what sounds like Wilson disease, and then the answer is **"slit-lamp exam."** - Can cause ↑ LFTs with cirrhosis and Parkinsonism. - Copper deposits in basal ganglia, especially the putamen. - Parkinsonism in a young patient = Wilson until proven otherwise. - In old patient, Parkinsonism = Parkinson disease, normal pressure hydrocephalus, Lewy-body dementia, or progressive supranuclear palsy. - Treat with the copper chelator penicillamine.
CFTR	- Cystic fibrosis; chromosome 7; autosomal recessive. - Codes for chloride channel. - In healthy patients, the *CFTR* chloride channel is normally located at cell surface. In cystic fibrosis, however, it remains sequestered in the rough endoplasmic reticulum / doesn't make it to the cell surface. - "Abnormal protein folding" is answer on NBME for result of *CFTR* mutations. - Delta F508 (ΔF508) is most common mutation, which is deletion of phenylalanine 508. - CF is textbook example of allelic heterogeneity, which means many different mutations can cause the same disease. For this reason, most genotyping panels lack sensitivity, and sweat-chloride test showing >60 mEq/L is most diagnostic/accurate. A nasal test showing increased potential difference can also be performed. - Presents as child with chronic history of lung infections. - *Pseudomonas* exceeds *S. aureus* after age 10. Students obsess over this age factoid, but what's more important is what the USMLE tells you in terms of the microbiology. If they say gram-negative rods, you know that's *Pseudomonas*; if they say gram-positive cocci in clusters, you know that's *S. aureus*. - Secretions in the alveoli and pancreatic ducts are inspissated (meaning desiccated / dried up within a lumen), making them sticky. This leads to exocrine pancreatic insufficiency, where enzymes can't make it to the duodenum → fat-soluble vitamin malabsorption → NBME exams love vitamin E deficiency in CF in particular (presents as neuropathy). - Meconium ileus at birth; buzzy, but often not mentioned in questions. - Phenotypically normal siblings of affected children have 2/3 chance of being carrier (holds true for any AR disorder). - Ivacaftor/lumacaftor are newer treatments. They are known as potentiators/correctors, where the Cl^- channel structure, location, and function are improved.
DMD	- XR disorder caused by mutation in dystrophin. - Mutation results in disruption of α-/β-dystroglycan, which is required for proper internal **cytoskeletal** anchoring of the muscle cell to the extracellular matrix. - Presents with pseudohypertrophy, where muscles appear large but are replaced with fibroadipose tissue (connective tissue stromal cells). - Duchenne presents in a young boy who implements Gower maneuver to stand up (uses arms to walk up off the floor because leg muscles are weak). - Becker presents in adolescence or young adulthood (less severe form of Duchenne). - Duchenne is classically frameshift mutation; Becker is classically not frameshift.

FBN1/2	- Marfan syndrome; autosomal dominant; chromosome 15. - Codes for fibrillin, which is a glycoprotein that forms a sheath around elastin. - Tall, lanky body habitus with chest wall abnormalities (i.e., pectus excavatum or carinatum), flat feet (pes planus), scoliosis, mitral valve prolapse (mid-systolic click), increased risk for aortic dissection (can retrograde propagate toward aortic root, causing root dilatation and aortic regurgitation [decrescendo diastolic murmur]). - Is not associated with berry/saccular aneurysms (unlike Ehlers-Danlos and ADPKD).
FMR1	- Fragile X; X-linked; caused by CGG trinucleotide repeat (TNR) expansion. - Results in **hypermethylation** of the gene and transcriptional silencing. - Presents as boy with large, everted ears; long, narrow jaw; macroorchidism; and intellectual disability. - Can sometimes be symptomatic in females if skewed X-inactivation (lyonization).
FXN	- Friedreich ataxia; Frataxin gene; GAA TNR expansion. - Presents with ataxia (you guessed it), scoliosis, cardiomyopathy, and decreased reflexes.
G6PD	- Glucose-6-phosphate dehydrogenase deficiency; X-linked recessive. - Presents as boy with hemolysis leading to jaundice (unconjugated hyperbilirubinemia) following oxidizing drug (e.g., sulfa, dapsone, primaquine) or viral illness. - Heinz bodies (denatured hemoglobin) and bite cells (degmacytes; partially phagocytosed RBCs due to removal of Heinz bodies) are seen.
HFE	- Hereditary hemochromatosis; autosomal recessive; chromosome 6. - Iron overload. - **Mechanism USMLE wants is "increased intestinal iron absorption."** - Body has poor ability to excrete iron; occurs naturally via menses in women; otherwise there are minor losses via skin shedding. - Main iron regulation is via shutting off intestinal absorption; this is impaired in hemochromatosis. - Usually presents in adulthood in males first (because of menses in women). - Can present as "bronze diabetes" → hyperpigmentation due to hemosiderin deposition in skin + ↑ fasting sugars (iron deposition in tail of pancreas). - Miscellaneous other findings can be seen like infertility (iron deposition in hypothalamus, anterior pituitary, or gonads), cardiomyopathy, or arthritis (pseudogout). - Hereditary hemochromatosis, primary hyperparathyroidism, and hypothyroidism are 3 most important causes of pseudogout. I used to only discuss the former two with students over the years, but the latter now shows up on a new 2CK NBME exam where they mention chondrocalcinosis (calcium deposition in cartilage, indicating pseudogout). - USMLE wants you to know there is ↑ risk of hepatocellular carcinoma. There is easy NBME Q of hemochromatosis where they ask what patient is at increased risk for, and the answer is simply "hepatocellular carcinoma." - Diagnose with **ferritin >300 mg/dL.** Transferrin saturation will also clearly be ↑. - It is exceedingly rare that ferritin is >300 in other conditions, however this can occur in lymphoma and leukemia, where ↑ ferritin is a poor prognostic marker in non-Hodgkin lympoma. There is one NBME Q on 2CK where ferritin is 300 where it's not hemochromatosis, but USMLE won't play gotchya. - Treat with **serial phlebotomy, not chelators.** - Chelators such as deferoxamine or deferasirox are for secondary hemochromatosis due to transfusional siderosis (i.e., repeated blood transfusions that contain iron, for e.g., β-thalassemia major).
HTT	- Huntington disease; autosomal dominant; CAG TNR expansion on chromosome 4. - Presents as cognitive decline and choreoathetosis, usually in patient 30s-40s. - Chorea = fast, purposeless, jerky movements. - Athetosis = slow, writing movements. - TNR disorders demonstrate **anticipation,** which means they become more severe and earlier-onset with each generation due to further expansion of the TNR, so the vignette can say the, e.g., 40-year-old patient had parent with similar symptoms appearing in his/her 50s. - If the USMLE asks you for which part of the brain is fucked up, choose caudate nucleus.
NFKB1/2	- Codes for NF-κB protein, which is involved in a myriad of cell-signaling processes.

	- For whatever magical reason, a repeated question across NBME exams wants you to know that "IκB releases NF-κB after undergoing phosphorylation." - NF-κB is then free to go to the nucleus to upregulate transcription of 150+ genes. - Glucocorticoids (i.e., such as prednisone) partially exert their immunosuppressant effects by inhibiting NF-κB-mediated gene expression.
PKD1/2	- Autosomal dominant polycystic kidney disease. - The answer on USMLE if disease starts as an adult (i.e., 30s-40s). - Cysts are technically present early in life, but only become clinical as adult (i.e., ↑ BP and ↑ RFTs). - These patients have ↑ BP due to ↑ RAAS (compression of microvasculature of kidney due to enlarging cysts). - Can cause saccular (berry) aneurysms of the circle of Willis → ↑ risk of subarachnoid hemorrhage. - Highest yield point is that **serial blood pressure checks** are correct over circle of Willis MR angiogram screening. Latter is wrong answer on USMLE. MR angiogram screening of circle of Willis is only done when there is (+) family Hx of SAH or saccular aneurysms. - Step 1 NBME gives easy vignette of ADPKD, and then the answer is just "polycystin" as the protein that's fucked up. Sounds obvious, but it's asked so I'm mentioning it. - Cystic kidneys are part of "ciliopathies," which is obscure term that refers to conditions where cilia are abnormal. Polycystin is a protein required for cilia function on renal epithelium. - Most common extra-renal location for cysts is the liver (85% by age 30).
PKHD1	- Autosomal recessive polycystic kidney disease. - The answer on USMLE for cystic kidneys in pediatrics. - Can be associated with **hepatic fibrosis.**

Notable tumor markers for USMLE	
AFP	- Alpha-fetoprotein. - Yolk sac tumor (aka endodermal sinus tumor) → the answer on USMLE for a testicular or ovarian tumor in a child. - Mixed germ cell tumors (if hCG also high). - Hepatocellular carcinoma (HCC) and hepadenomas can also have high AFP.
ALP	- Alkaline phosphatase. - Placental ALP can sometimes be increased in seminoma. - Unrelated to increased ALP in bile duct obstruction or bone fractures.
CA 15-3	- Breast cancer.
CA 19-9	- Pancreatic cancer.
Calcitonin	- Increased in medullary thyroid carcinoma. - Normal role of calcitonin is to inhibit osteoclast activity.
CEA	- Carcinoembryonic antigen. - Colon cancer (although non-specific).
hCG	- Human chorionic gonadotropin. - Choriocarcinoma. - Mixed germ cell tumors (if AFP also high).
LDH	- Lactate dehydrogenase. - Dysgerminoma.
PSA	- Prostate-specific antigen. - Prostate cancer.

HY autoantibodies for USMLE	
HY disease association	**HY antibodies**
Antiphospholipid syndrome	- Anti-β2-microglobulin; anti-cardiolipin; lupus anticoagulant. - The latter is the name for either of the former two if the patient happens to have SLE. - Presents as *in vivo* thromboses despite an ↑ *in vitro* aPTT. - These antibodies are associated with recurrent miscarriage. - Can cause false-positive syphilis screening (i.e., SLE patient who gets positive syphilis VDRL test).
Bullous pemphigoid	- Anti-hemidesmosome (bullous pemphigoid antigen). - Hemidesmosomes connect the dermis to epidermis at the basement membrane. - In BP we see linear immunofluorescence on skin biopsy because hemidesmosomes line the basement membrane. - Q can say there are sub-epidermal blisters (pemphigus vulgaris, in contrast, the Q can say intra-epidermal blisters). - Usually no oral involvement, but not mandatory rule. - No Nikolsky sign (i.e., no sloughing of skin with friction).
Pemphigus vulgaris	- Anti-desmosome (anti-desmoglein 1 and 3). - Desmosomes connect adjacent epidermal cells side-to-side. - In PV we see a net-like immunofluorescence on skin biopsy because the desmosomes are between / outline the skin cells. - Q can say there are intra-epidermal blisters (bullous pemphigoid, in contrast, the Q can say sub-epidermal blisters). - Can present with oral involvement, albeit not mandatory. - (+) Nikolsky sign (i.e., sloughing of skin with friction).
Goodpasture syndrome	- Anti-collagen IV (anti-GBM; glomerular basement membrane). - Goodpasture presents as male 20s-40s with hematuria + hemoptysis + anti-collagen IV antibodies. - Causes linear immunofluorescence on renal biopsy, since collagen IV lines basement membranes. - Don't confuse with Alport syndrome, which is XR disorder due to *mutations* in type IV collagen (not Ab). - Alport presents as male with red urine + eye/ear problem.
Granulomatosis with polyangiitis (Wegener)	- c-ANCA, aka anti-proteinase 3 (anti-PR3). - Dumb mnemonic I created that helps some of my students: **W**ater **C**loset (**W**egener **C**-ANCA). - Presents are hematuria + hemoptysis + "head-itis" (i.e., problem with the head, e.g., nasal septal perforation, mastoiditis, sinusitis, necrotizing otitis). - Sometimes has neuro findings (mononeuritis multiplex).
Eosinophilic granulomatosis with polyangiitis (Churg-Strauss)	- p-ANCA, aka anti-myeloperoxidase (anti-MPO). - Presents as asthma + eosinophilia. - Sometimes has neuro findings (mononeuritis multiplex).
Microscopic polyangiitis	- p-ANCA (anti-myeloperoxidase). - Presents as usually just red urine in person with (+) p-ANCA. - Sometimes has neuro findings (mononeuritis multiplex).
Systemic lupus erythematosus (SLE)	- Anti-double-stranded DNA (dsDNA). - Anti-Smith (ribonucleoprotein). - Anti-hematologic cell line Abs. - Lupus anticoagulant. - Should be noted that anti-dsDNA goes up in acute flares + best reflects renal prognosis.

	- Anti-Smith is *more specific* than anti-dsDNA, meaning it shows up less often than anti-dsDNA, but if it does show up, the patient absolutely is ruled-in for having SLE. - Thrombocytopenia is an exceedingly HY finding in SLE due to the anti-hematologic cell line Abs. - If all cell lines are down in SLE, aplastic anemia (i.e., ↓ bone marrow production) is wrong answer; choose "increased peripheral destruction" as answer instead, since the cause is merely antibodies against the cells. - Lupus anticoagulant I discussed above in anti-phospholipid syndrome; causes recurrent miscarriages and false (+) syphilis tests. - Most common presenting feature in SLE overall is arthritis. The Q need not mention malar rash (I'd say only ~1/3 of Qs). So if you get, e.g., 34-year-old woman with arthritis and thrombocytopenia + no other info, that is pass-level SLE till proven otherwise. - Complement protein C3 can be down in SLE flares. - Renal diagnosis for lupus nephritis is basically always diffuse proliferative glomerulonephritis (DPGN) on NBME.
Drug-induced lupus (DIL)	- Anti-histone antibodies. - Caused by various drugs (**M**om is **HIPP**): **M**inocycline, **H**ydralazine, **I**NH, **P**rocainamide, **P**enicillamine. - Presents usually as arthritis + pericarditis or mediastinitis in patient who was given one of the above buzzy meds.
Myasthenia gravis	- Post-synaptic nicotinic acetylcholine receptor antibodies. - Sometimes a paraneoplastic of thymoma, but need not be. - Presents usually as 40s woman with triad of diplopia, eyelid ptosis, and dysphagia or hoarseness of voice. Gets worse with activity, whereas Lambert-Eaton improves. - USMLE wants pyridostigmine (cholinesterase inhibitor) as Tx.
Lambert-Eaton	- Presynaptic voltage-gated calcium channel antibodies. - Sometimes a paraneoplastic of small cell lung cancer. - Presents as proximal muscle weakness that improves with activity (e.g., patient can stand up from chair after trying a few times).
Small cell cerebellar dysfunction	- Anti-Hu/-Yo. - Presents as ataxia in someone with small cell lung cancer and negative CNS imaging (meaning metastases to brain aren't the cause of the neurologic issues).
Polymyositis / Dermatomyositis	- Anti-Jo1. - Polymyositis and dermatomyositis are usually idiopathic autoimmune conditions, but they can sometimes be a paraneoplastic of ovarian cancer. - Present as proximal muscle weakness on physical examination + increased serum CK. - Dermatomyositis = polymyositis + skin findings. The latter are shawl rash, Gottron papules, mechanics hands, and heliotrope rash.
Limited-type systemic sclerosis / scleroderma	- Anti-centromere. - Scleroderma = another name for systemic sclerosis. - Limited type merely = CREST syndrome = Calcinosis, Raynaud phenomenon, Esophageal dysmotility, Sclerodactyly, Telangiectasias. - Pulmonary fibrosis leading to pulmonary HTN is HY for both limited and diffuse types of scleroderma.

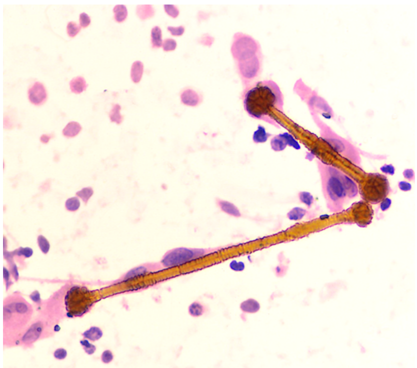
Diffuse-type systemic sclerosis / scleroderma	- Anti-topoisomerase I (Scl-70). - Can present as same findings as in limited-type, but also with malignant HTN (e.g., BP of 220/120) due to renal involvement causing a surge of RAAS.
Primary biliary cirrhosis	- Anti-mitochondrial. - Presents as woman 20s-50s with ↑ ALP, direct bilirubin, and serum cholesterol, with generalized pruritis and Hx of other autoimmune disease in her or family member (e.g., RA, SLE).
Autoimmune hepatitis	- Anti-smooth muscle. - Presents as woman 20s-30s with idiopathic elevation of hepatic transaminases. - Diagnosis of exclusion (meaning you have to eliminate the other answer choices to get there).
Rheumatoid arthritis	- Rheumatoid factor (an IgM Ab against the Fc region of IgG). - Anti-cyclic citrullinated peptide (CCP) antibodies. - Anti-CCP is *more specific* than rheumatoid factor for RA.
Sjögren syndrome	- Anti-SS-A (anti-Ro); anti-SS-B (anti-La). - Presents as xerostomia (dry mouth) + xerophthalmia (dry eyes) + arthritis.
Graves disease	- Anti-TSH receptor; this antibody is called thyroid-stimulating immunoglobulin, or TSI, and *activates* the TSH receptor. - Most common cause of hyperthyroidism. - Exophthalmos and pretibial myxedema are HY findings.
Hashimoto thyroiditis	- Anti-thyroperoxidase, aka anti-microsomal antibodies. - Anti-thyroglobulin. - Highest yield cause of hypothyroidism in Western countries. - Worldwide the most common cause is iodine deficiency. - Buzzy findings are weight gain, brittle hair, dry/doughy skin, and cold intolerance. The USMLE often will omit these findings because they're too easy. - HY additional findings I've seen show up in NBME Qs are: bradycardia (55-60), menstrual irregularity, dysthymia/depression/apathy, carpal tunnel syndrome (due to glycosaminoglycan [GAG] deposition), ↑ cholesterol, proximal muscle weakness with ↑ creatine kinase (hypothyroid myopathy), transaminitis (sounds weird, but LFTs can be ↑). - Lymphocytic infiltrate is seen on Hashimoto thyroid biopsy.
Pernicious anemia	- Anti-parietal cell or anti-intrinsic factor antibodies. - Presents as B12 deficiency (intrinsic factor is needed to bind to B12 so it can be absorbed at the terminal ileum).
Celiac disease	- Anti-endomysial (aka anti-gliadin). - Anti-tissue transglutaminase IgA (antibody screening will yield false-negatives if patient also has IgA deficiency). - Insensitivity to wheat, oats, rye, and barley. - Duodenal biopsy shows flattening of the intestinal villi on biopsy. Can present with iron deficiency anemia.
Type I diabetes mellitus	- Anti-glutamic acid decarboxylase (anti-GAD). - Anti-zinc transporter 8.
Primary membranous glomerulonephritis	- Anti-phospholipase A2 receptor. - Membranous glomerulonephritis is a nephrotic syndrome. - Other HY causes of MG are drugs (gold salts, dapsone, sulfonamides), autoimmune diseases like RA, cancer (e.g., pancreatic, breast), and hepatitis B. So any of these patients + nephrotic syndrome = MG usually.
Thrombotic thrombocytopenic purpura (TTP)	- Anti-ADAMTS13, which is a matrix metalloproteinase that cleaves vWF multimers.

	- Presents as pentad of 1) thrombocytopenia, 2) schistocytosis (the combo of 1+2 is called microangiopathic hemolytic anemia; MAHA), 3) renal issues (hematuria or ↑ RFTs), 4) fever, 5) neurologic signs.
Immune thrombocytopenic purpura (aka idiopathic thrombocytopenic purpura; ITP)	- Anti-GpIIb/IIIa on platelets (platelet aggregation). - Presents classically as school-age kid who gets petechiae and/or nose bleeds following a viral infection. - Can also present in random woman 30s-40s with ↑ bleeding time and ↓ platelet count. - The Q need not mention the viral infection, but if they do, then the Q is hyper easy and pass-level.
Heparin-induced thrombocytopenia (HIT)	- Antibodies against platelet factor 4-heparin complex (i.e., anti-PF4-heparin). - Presents as (you'd never guess) thrombocytopenia following heparin administration (e.g., for DVT or PE). - Treatment is direct thrombin inhibitor (e.g., dabigatran); warfarin is wrong answer for Tx.
Addison disease	- Can sometimes be caused by anti-21-hydroxylase Abs. - Low Na^+, high K^+, low bicarb, low pH, high ACTH. - Hyperpigmentation common.
Rheumatic heart disease	- Anti-myosin; anti-valve-derived proteins. - Should be noted that these Abs are formed against Group-A Strep (*S. pyogenes*) M protein and almost always cross-react with the mitral valve (molecular mimicry). - Type II hypersensitivity. Don't confuse with PSGN, which is type-III hypersensitivity. - Patient will have mitral regurg early but mitral stenosis late. - Sydenham chorea is autoimmune condition of basal ganglia. - Erythema marginatum is characteristic skin condition.
Neuromyelitis optica (Devic syndrome)	- Anti-aquaporin 4 (asked on 2CK Neuro form). - Presents basically the same as multiple sclerosis except they'll tell you these Abs are positive and that the CSF is negative for IgG oligoclonal bands (normally seen in MS).
Cold autoimmune hemolytic anemia	- IgM against RBCs (MMM ice cream; since ice cream is cold). - CMV and *Mycoplasma* are HY infectious associations. - (+) Coombs test (just means antibodies against RBCs).
Warm autoimmune hemolytic anemia	- IgG against RBCs. - Caused by various miscellaneous drugs and infections. - Associated with chronic lymphocytic leukemia (CLL). - (+) Coombs test (just means antibodies against RBCs).

Hypersensitivity types for USMLE

Type	Mechanism
I	- Immediate (within minutes): Antigen binds to Fab region of IgE on the surface of mast cell and basophils. This causes adjacent IgE crosslink on the cell surface to crosslink. The mast cell / basophil then degranulates and secretes histamine + tryptase. - Late (within hours): Cytokines recruit inflammatory cells (e.g., eosinophils) → more inflammation. - HY examples are anaphylaxis (i.e., bee sting, peanut allergy), atopy / asthma.
II	- Antibody production against one's own cells, tissues, receptors. - HY examples are Graves disease, Hashimoto thyroiditis, Goodpasture syndrome, pernicious anemia, HIT, ITP, rheumatic fever, myasthenia gravis, Lambert-Eaton, bullous pemphigoid, pemphigus vulgaris.
III	- Antibody-antigen complexes (i.e., antibodies simply bind to antigen and deposit; the antibodies do not target cells, tissues, or receptors).

	- HY examples are post-streptococcal glomerulonephritis, polyarteritis nodosa, SLE, arthus reaction, serum sickness.
IV	- T cell response (only HS type not associated with antibodies); can be mediated by either CD8+ or CD4+ T cells. - HY examples are PPD skin test for TB, contact dermatitis (poison ivy/oak/sumac, sunscreen, nickel on watches, latex), graft-vs-host disease.

Notable Carcinogens for USMLE	
Alcohol	- Oral/pharyngeal SCC; SCC of esophagus; hepatocellular carcinoma; breast cancer.
Aflatoxin	- Answer on USMLE if they mention peanut farming in China. - Can cause hepatocellular carcinoma. On one offline Step 1 NBME.
Aniline dyes	- Answer on USMLE for hematuria in textile factory worker. - Can cause transitional cell carcinoma (TCC) of the bladder. - Aniline dyes are a type of industrial clothing dye.
Arsenic	- Answer on USMLE for Mees lines (white lines on nails), or palms + soles rash, in a gardener. - Arsenic is in many fertilizers; a small amount causes plants to flourish; poisoning seen in gardeners/landscapers, clearly.  - An NBME Q gives pic of Mees lines in gardener → answer = arsenic.
Asbestos	- Asbestosis → lung disease (usually restrictive) with ferruginous bodies and supradiaphragmatic / pleural plaques. - NBME Q shows pic of ferruginous body and asks what cell initiates pulmonary fibrosis → answer = macrophage.  (Ferruginous bodies in asbestosis) - The pleural / supradiaphragmatic plaques can be described simply as “soft tissue densities” seen on CXR. - Can develop into mesothelioma → can appear as white cancer that circumferentially envelops the lungs; NBME Q asks for “mesothelial cells” as the answer. - Shipyard workers, construction workers, and electricians are buzzy professions on USMLE for asbestosis.
Benzene	- Leukemias.

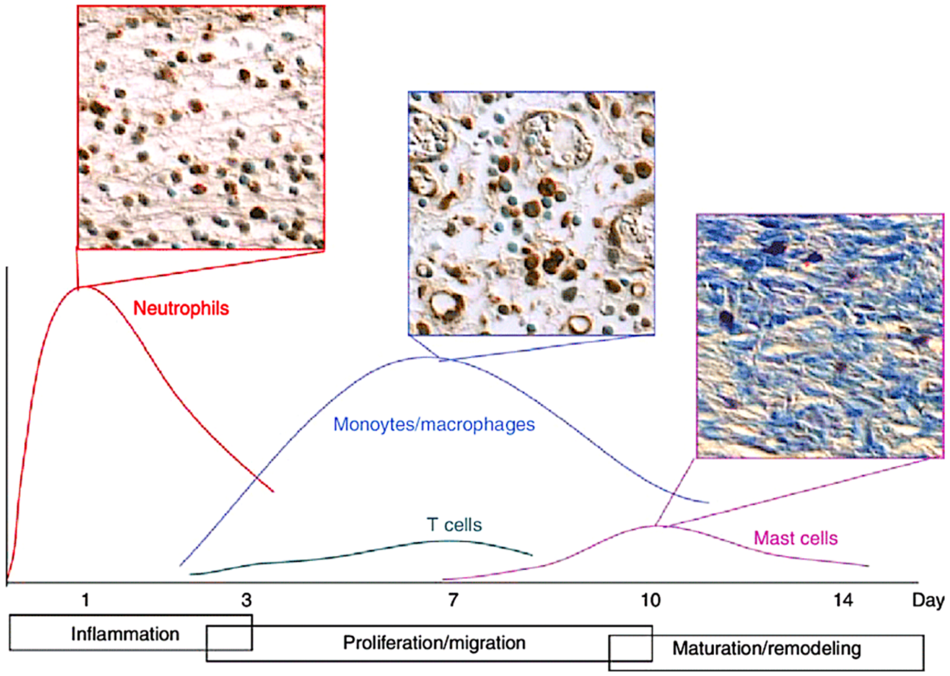
Ethanol	- Highest yield = esophageal squamous cell carcinoma + hepatocellular carcinoma. - Breast cancer carries theoretical association but USMLE doesn't give a fuck.
Ionizing radiation	- Thyroid cancer; leukemias.
2-naphthylamine	- Used to be used in some industrial dyes. Use discontinued to due carcinogen risk. - Causes collecting duct cancer (on NBME), and urothelial cancer in general.
Nitrosamines	- Gastric cancer. - Associated with consumption of smoked / cured meats.
Processed meats	- Colorectal cancer.
Radon	- Adenocarcinoma of the lung in non-smokers. - Due to background radiation from the ground; considered the second-most important cause of lung cancer after smoking.
Smoking/tobacco	- Basically every cancer.
UV light	- Skin cancer.
Vinyl chloride	- Hepatic angiosarcoma. - Can appear as nodular lesions of the liver (tricky, because can resemble metastases).

Notable Paraneoplastic syndromes for USMLE	
Gastric	- Acanthosis nigricans. Presents as velvety dark skin around the base of the neck / on the shoulders. Should be noted that almost always this finding simply means insulin resistance, but just be aware it can also be associated with visceral malignancies like gastric. "Visceral malignancy" is a general term that refers to main organ system cancers like the liver, pancreas, stomach, etc. - Sign of Leser-Trelat (sudden appearance of large numbers of seborrheic keratoses). Can occur with visceral malignancies, like gastric and pancreatic. - Virchow node (Troisier sign) is a palpable left supraclavicular lymph node that is classically associated with gastric cancer, but can be visceral malignancy in general. There is also an NBME Q where Virchow node presents on the right side in a Hodgkin patient.
Ovarian	- Can cause dermatomyositis. Weird, but asked on an NBME question.
Pancreatic	- Migratory thrombophlebitis (Trousseau sign of malignancy). - Presents as random veins that become swollen, red, and thrombosed. - Not limited to pancreatic cancers; tends to be associated with adenocarcinomas in general. But pancreatic adenocarcinoma is the classic association.
Renal cell carcinoma	- Hypercalcemia (PTHrp secretion, same as squamous cell of lung). - Polycythemia (EPO secretion). - Q will give you smoker with hypercalcemia, polycythemia, and red urine + they can show image of large clear cells (HY histo; RCC is aka clear cell carcinoma).
Lung cancer (general)	- Hypertrophic osteoarthropathy (clubbing + periostitis). - Periostitis is inflammation of the periosteum over the bone. - Classically adenocarcinoma of the lung, but the USMLE won't specify. - They'll just give you vignette of clubbing + hand pain, and the answer is "chest x-ray" for next best step in management.
Neuroblastoma	- Opsoclonus-myoclonus syndrome (dancing eyes).
Small cell carcinoma of the lung	- Cushing syndrome (ACTH secretion). - SIADH (ADH secretion). - Lambert-Eaton syndrome (Abs against presynaptic voltage-gated Ca^{2+} channels). - Cerebellar dysfunction/ataxia (anti-Hu/-Yo antibodies).
Squamous cell carcinoma of the lung	- Hypercalcemia/hypophosphatemia (PTHrp secretion). - PTHrp is not the the same as endogenous PTH, which is suppressed due to negative-feedback from high calcium.
Thymoma	- Myasthenia gravis (Abs against postsynaptic nicotinic acetylcholine receptors). - Do CXR / chest CT in patients with MG. - Thymoma can also rarely cause pure-RBC aplasia (i.e., only RBCs are down).

Cancer Stage vs Grade	
Stage	**- How far the tumor has spread.** - TNM system = Tumor size/invasiveness; Lymph nodes; Metastases. - Cancer that is *in situ* (i.e., confined superficial to basement membrane) is Stage 0. - NBME likes the term "microinvasion," which means "crossed the basement membrane." - If you're asked about melanoma + invasion to which depth is associated with worst prognosis, answer = "subcutaneous tissue" or "hypodermis," not basement membrane. Even though crossing the BM is HY for microinvasion, which is when stage has officially increased, prognosis with skin cancer is directly related to depth of the lesion. - Cancer that has metastasized to distant organs is highest stage.
Grade	- **# of mitoses + degree of anaplasia**. - Don't confuse anaplasia with aplasia. Anaplasia means the degree of undifferentiation / immaturity of the cells. Aplasia means lack of growth. If a skin cell, for instance, looks nothing like it's supposed to and instead resembles a primitive stem cell, then it has a high degree of anaplasia (i.e., it has reverted to a more immature / undifferentiated state). - High number of nucleoli = high protein synthesis = high # of mitoses. - 1:1 nuclear:cytoplasmic ratio → nucleus is huge → high # of mitoses. - ↑ Ki-67 staining = high grade. Ki-67 is a protein expressed in actively dividing cells.
- For USMLE purposes, stage is more important than grade for prognosis. This is exceedingly HY and prevalent on NBMEs. Some students get pedantic about grade being more important than stage for some brain tumors. I have never once seen this notion assessed on NBMEs and risks getting you questions wrong. - For example, an *in situ* cancer that demonstrates aggressive mitoses and has cells poorly resembling the native tissue = low stage, high grade. - A cancer that has metastasized but has cells that are slowly dividing and well-differentiated = high stage, low grade. - The way this applies on USMLE is they will ask for which of the following is most likely to indicate the need for cystectomy in a patient (i.e., their way of asking which has the worst prognosis), and the answer will be something like, "tumor cells present in sentinel lymph node that are well-differentiated and very slowly dividing." The student thinks this doesn't sound that bad, but it will be the only answer where stage has increased. All of the other answers might give *in situ* cancers with high grade, so they sound more sinister, but are wrong.	

Locations of metastases	
Metastasizes to	**Cancer type**
Spine/vertebrae	- Prostate, breast, and lung cancer all love to metastasize to spine/vertebrae. I've seen this across NBME Qs (and especially on 2CK Neuro CMS Qs), where they give neurologic findings in patients with cancer and want "epidural spinal cord metastases," or "epidural spinal cord compression," or "metastases to cauda equina" as answers. - There is a Step 1 NBME Q where they give lytic lesions of the vertebrae in a vague 1-2-liner, where multiple myeloma isn't listed, and they just want "metastatic breast cancer" as the answer. - One of the highest yield general principles for USMLE is that prostate cancer causes **osteoblastic** metastases, which means they light up on a bone scan. Pretty much all other cancers cause osteolytic metastases. Occasionally some cancers such as breast can cause osteoblastic metastases, but for USMLE this application is essentially nonexistent; just remember prostate for this point.
Lung/brain	- Choriocarcinoma loves to go to lung and brain. If they give gynecologic question + mention pulmonary nodules or stroke-like presentation, this is HY for choriocarcinoma. Hydatidiform mole is a wrong answer here because, even though the latter can progress to invasive mole and choriocarcinoma, if the Q itself already mentions pulmonary or neuro findings, we know we already have the cancer, so choriocarcinoma is the better answer.
Ovaries	- Gastric cancer can metastasize hematogenously to ovaries. These are called Krukenberg tumors and will have signet ring cells on biopsy, which contain mucin.

Omentum	- USMLE loves direct extension to the omentum for ovarian cancer. There are two questions on the NBMEs where they say “omental thickening” and “omental caking.”
Para-aortic lymph nodes	- Testicular and ovarian cancers. - I discuss specific lymph node drainages of cancers in my HY Immuno PDF, but this one is most important, so I’m including it here.

Cells of inflammation / approximate timelines	
Neutrophils	- **The cell of acute inflammation.** - Neutrophils are “first responders” at sites of inflammation and are rapidly recruited to the site of injury or infection, where they play a critical role in the initial response to pathogens or tissue damage within the first 24-48 hours. - Highly phagocytic cells that engulf and kill microorganisms, foreign antigens, and cellular debris through. - Carry out respiratory burst. Phagocytosed microbes are killed via NADPH oxidase used to generate H_2O_2, and via myeloperoxidase to produce bleach. - Complement protein C5a, IL-8, LTB-4, kallikrein, and platelet-activating factor are pro-chemotaxis that stimulate/recruit neutrophils to sites of infection and inflammation. - Pus is an accumulation of dead neutrophils, cellular debris, pathogens, and tissue fluids at the site of infection or inflammation. - Normally ~50-60% of WBCs on USMLE labs. If > 60%, we call this neutrophilic shift, which suggests bacterial infection. “Bands” are immature neutrophils that are sometimes increased in bacterial infections. - Neutrophilia (high neutrophils) seen where total WBC count is >30,000/µL, we call this “reactive granulocytosis,” which is aka leukemoid reaction, or “increased leukocyte release from bone marrow post-mitotic reserve pool.” WBC count should normally be 4-11,000. If bacterial pneumonia or ruptured appendix, for instance, we expect WBCs to maybe be 12-20,000s, but >30k, it’s leukemoid reaction. “-Oid” means looks like but ain’t, so it looks like leukemia because the WBC is so high, but rather than leukemic cells on smears, we see neutrophils instead. I discuss this stuff in more detail in the HY Heme/Onc PDF. - Neutropenia (aka agranulocytosis) is seen with chemo-/radiotherapy, viral infections (e.g., Parvo B19), and drugs (e.g., clozapine, methotrexate, carbamazepine). 

Macrophages	- **The cell of chronic Inflammation.**
	- Following neutrophils, they begin predominance of migration into tissues after 48 hours. There is an NBME Q, for example, where they say there is a scarred segment of Fallopian tube in chronic pelvic inflammatory disease (PID), and they ask which cell is most likely to predominate on biopsy, and the answer is macrophage. Neutrophil is wrong because they specify chronic.
	- Antigen-presenting cells (APCs). Phagocytose antigens and present it them to CD4+ T cells, initiating adaptive immune responses.
	- Produce cytokines that carry out a variety of effects. TNF-α causes increased vascular permeability and swelling. IL-1 causes fever. IL-6 causes acute-phase protein release by the liver (e.g., CRP). IL-12 stimulates CD4+ T cells.
	- Play a role in septic shock, both due to endotoxin as well as superantigen.
	    - Shock due to endotoxin: Lipid A of lipopolysaccharide (LPS) binds CD14 (Toll-like receptor; TLR) on macrophages, leading to release of cytokines, namely TNF-α (hypotension; vasodilation; vascular permeability) and IL-1 (fever).
	    - Shock due to superantigen: toxic shock syndrome toxin (TSST) of *S. aureus* or exotoxin A of *S. pyogenes* cause toxic shock syndrome and toxic shock-like syndrome, respectively. Both toxins bridge MHC-II on the macrophage with T cell receptor (TCR) on CD4+ T cells, causing release of cytokines from macrophages.
	- Can organize into granulomas, which are aggregations of macrophages that fuse to form a multi-nucleated collection (i.e., granuloma). Activated macrophages are aka epithelioid macrophages, or histiocytes. These can be present idiopathically/miscellaneously in chronic inflammation (e.g., Crohn, sarcoidosis), or can attempt to phagocytose and digest foreign antigen over the long-term (e.g., remnant suture material for suture granulomas; ferruginous bodies in asbestosis).
	- Non-caseating granulomas in sarcoidosis secrete 1α-hydroxylase, leading to hypercalcemia (more detail in HY Arrows PDF). Crohn is other HY condition with non-caseating granulomas.
	- Caseating granulomas are seen in TB, fungal infections, and cat scratch disease, as mentioned earlier.
	- Play a role in pathogenesis of atherosclerosis – i.e., phagocytose oxidized lipid particles (more detail in HY Cardio PDF).
	- Tissue repair → promote formation of granulation tissue and fibrosis. They release growth factors (discussed in below table) that stimulate fibroblasts and endothelial cells, aiding in wound healing.
	- Reside in specific tissues with unique names, such as Kupffer cells in the liver, microglia in the central nervous system, and alveolar macrophages in the lungs. These tissue-resident macrophages have specialized functions and contribute to local immune surveillance.

Growth factors released by macrophages during inflammation	
TGF-β	- Transforming growth factor-β. - Stimulates the proliferation and activation of fibroblasts and myofibroblasts, promoting collagen production and tissue repair; also plays a role in angiogenesis by stimulating endothelial cell growth. - USMLE will give you a Q where they ask which cytokine helps with extracellular matrix homeostasis, then you just eliminate to get there.
PDGF	- Platelet-derived growth factor. - Recruits and activates fibroblasts; stimulates smooth muscle cells and endothelial cells, contributing to blood vessel formation.
VEGF	- Vascular Endothelial Growth Factor. - Stimulates angiogenesis.

	- USMLE wants you to know that in tumorigenesis, VEGF synthesis is stimulated by localized ↓ **pO2**. In other words, blood supply to a tumor is the same initially, but we now have more cells, therefore more oxygen demand, so pO2 at the site of the tumor falls. This causes VEGF to ↑ → angiogenesis → tumor now has adequate blood supply.
FGF	- Fibroblast Growth Factor. - Stimulates fibroblast proliferation, collagen synthesis, and tissue remodeling.
EGF	- Epidermal Growth Factor - Promotes the proliferation and migration of epithelial cells, which is required for the re-epithelialization and the closure of wounds.

Wound healing HY Phases	
Phase 1: Hemostasis	- Occurs immediately following tissue injury. - Involves vasoconstriction to reduce bleeding and the formation of clot or hemostatic plug to seal the a wound. - Platelet activation and coagulation cascade ensue. - An NBME Q has "increased endothelin at site of wound" as a correct answer for a guy who cuts his neck. You need to know endothelin is a vasoconstrictor that is not limited to the lungs.
Phase 2: Inflammatory	- First ~3 days. - Extravasation of WBCs; neutrophils are predominant cell initially, with macrophages entering after a couple days. - Redness (rubor), heat (calor), swelling (tumor), and pain (dolor).
Phase 3: Proliferative	- Days to weeks. - Granulation tissue formation (pink, vascular tissue rich in type III collagen); angiogenesis; epithelial cell proliferation; myofibroblast-mediated wound contraction.
Phase 4: Remodeling	- Weeks to months. - Type III collagen (pink; weaker) is replaced with type I collagen (white; stronger); 80% of tensile strength is restored in scars; tensile on USMLE = collagen if they ask for the extracellular matrix protein. - TGF-β is important for increasing fibroblast migration and proliferation, increased synthesis of collagen and fibronectin, and decreased degradation of extracellular matrix by metalloproteinases (all of this is directly from NBME stem). - Myofibroblast and matrix metalloproteinases are the answers if they ask about scar contraction. - Macrophage is the answer if they ask for the cell that initiates fibrosis.

Leukocyte extravasation + chemotaxis HY points, sans the superfluousness	
Step 1: Margination + rolling	- Carried out by E-, P-and L-selectins on endothelial cells that bind to WBCs.
Step 2: Tight binding	- LFA-1/CD-18 integrin is needed; deficient in leukocyte adhesion deficiency.
Step 3: Diapedesis	- Carried out by PECAM-1.
Step 4: Migration	- Carried out by LTB-4, IL-8, C5a, kallikrein, platelet-activating factor, and bacterial products.

Malignant tumor nomenclature	
- The word malignant means "has metastatic potential."	
Carcinoma	- Type of cancer that originates from epithelial cells (e.g.., skin, breast, prostate, GIT). - Adeno is a prefix that means "glandular" / "glands." So an adenocarcinoma is a type of carcinoma that originates from **glandular** epithelial cells (e.g., gastric adenocarcinoma → cancer of stomach glands; prostatic adenocarcinoma → cancer of prostate gland).

	- An NBME Q, for example, gives a biopsy that shows cells where morphology is neither squamous nor glandular in origin, and the answer is small cell. I talk about this in the HY Pulm PDF, but the point is, adenocarcinoma is wrong because they say "not glandular."
Sarcoma	- Type of cancer that originates from connective tissues (i.e., bones, muscles, cartilage, blood vessels). - In contrast to carcinomas, which arise from epithelial cells, sarcomas are characterized by malignant transformation of mesenchymal cells. - Mesenchymal cells are a type of multipotent stem cell found in connective tissues and have the capacity to differentiate into many different cell types – i.e., osteoblasts, chondrocytes, adipocytes, and vascular smooth muscle cells. - HY examples are osteosarcoma, chondrosarcoma, and liposarcoma.
Blastoma	- Malignant tumors that originate from cells that have not fully matured or differentiated. - HY examples are neuroblastoma, hemangioblastoma, glioblastoma multiforme, retinoblastoma, nephroblastoma (aka Wilms tumor).
Glioma	- Malignant tumor of glial cells (non-neuronal cells in CNS that provide support and maintenance functions for neurons). - HY examples are oligodendroglioma, glioblastoma multiforme (GM), and astrocytomas. You should know for USMLE that GM is a "grade IV fibrillary astrocytoma" (on NBME).
Myeloma	- Malignancies originating from plasma cells. - HY example is multiple myeloma. - Another example is plasmacytoma, which is a collection of malignant plasma cells.
Melanoma	- Malignant tumor of melanocytes.
Lymphoma	- Malignant tumor of the lymphatic system (i.e., Hodgkin, non-Hodgkin).
Leukemia	- Malignancy of leukocytes (white blood cells).
Benign tumor nomenclature	
-Oma	- Refers to benign tumors. - HY examples are adenoma (benign tumor of glandular tissue), lipoma (benign tumor of adipocytes), meningioma, rhabdomyoma (benign tumor of striated muscle), leiomyoma (tumor of smooth muscle, which will be uterine fibroids on USMLE), and thyroid toxic adenomas (secrete thyroid hormone).

Malignancy-associated microbes	
Aspergillus	- Fungus. - Can produce aflatoxin, which increases risk of hepatocellular carcinoma.
Clonorchis sinensis	- Trematode (fluke); a type of helminth (parasitic worm). - Can cause cholangiocarcinoma (cancer of the bile ducts). - Treat with praziquantel.
EBV	- Ebstein-Barr virus is part of Herpesviridae (DNA; enveloped; ds-linear). - Causes nasopharyngeal carcinoma and B-cell lymphomas (Hodgkin and NHL).
Helicobacter pylori	- Spiral-shaped gram-negative rod. - Causes mucosa-associated lymphoid tissue (MALT) lymphoma, a type of B-cell lymphoma. - I discuss *H. pylori* in detail in the HY Gastro PDF.
HepB/C	- Hepatocellular carcinoma
HHV-8	- Human herpesvirus 8 (aka Kaposi sarcoma-associated herpesvirus). - Kaposi sarcoma.
HIV	- Human immunodeficiency virus. - Primary CNS lymphoma at CD4 counts generally <100/μL. - Immunodeficiency in general can increase risk of squamous cell carcinomas (e.g., anal SCC in MSM; esophageal SCC in smoking/alcohol; SCC of skin due to sunlight); NBME Qs can occasionally mention immunodeficiency in Qs to imply SCC.
HPV 16/18	- Human papillomavirus. - Squamous cell carcinoma (usually genital/anal).

	- Yes there are other strains, but chill. USMLE cares about 16/18, not the other ones you can run off.
HTLV	- Human T-cell lymphotropic virus. - Can cause cutaneous T-cell lymphoma (mycosis fungoides) and T-cell leukemia (Sezary syndrome).
Schistosoma hematobium	- Trematode. - Squamous cell carcinoma of the bladder (patient who swam in lake in Africa). - Don't confuse with transitional cell carcinoma of bladder, which is most frequently due to smoking; TCC can also be caused by aniline (industrial) dyes, and 2-naphthylamine (another type of industrial dye).
Strep bovis	- Can increase risk of colorectal cancer (CRC). - Can also cause endocarditis in setting of CRC.

Histochemical stains	
- Idea isn't to be exhaustive/superfluous here. Point is to stay HY.	
Acid-fast	- Mycobacterium; *Cryptosporidium parvum*.
Desmin	- Muscle cells.
Giemsa	- *Plasmodium* (malaria); *Trypanosoma* (sleeping sicknesses and Chagas disease); *Leishmania*.
H&E	- Hemotoxylin and Eosin. - Cell nuclei stain blue (basophilic); cytoplasm and ECM are pink (eosinophilic).
India ink; mucicarmine	- *Cryptococcus neoformans.* - India ink is black background with white halo'ed yeasts; mucicarmine is red stain.
Oil Red O; Sudan black	- Fat. - I've seen these show up in fat embolism Qs, and also for fecal staining in steatorrhea.
PAS	- Periodic Acid-Schiff. - Differentiates glycogen (PAS-positive) from mucin (PAS-resistant). Helps distinguish glycogen-rich tumors from mucin-rich tissues or tumors. - Only thing you need to know for USMLE is Whipple disease = "PAS-positive macrophages in the lamina propria."
Prussian blue	- Stains iron. - Sideroblastic anemia (ringed sideroblasts are RBC precursors that contain excess iron deposits in their mitochondria; this image is buzzy for sideroblastic anemia). I talk about this stuff in detail in my HY Heme/Onc PDF.
Silver	- *H. pylori*, spirochetes (i.e., *Treponema pallidum, Borrelia*), *Leishmania, Pneumocystis.*

Collagen disorders	
Type I	- Found in bone; predominates in late wound healing (white in color); ↑ tensile strength than type III. - Osteogenesis imperfecta → fractures at different stages of healing; often mistaken for child abuse; blue sclerae (too easy; often omitted from Qs); conductive hearing loss (malformation of ossicles); if ruled out OI + child abuse, think osteopetrosis.
Type II	- Found in cartilage, intervertebral discs, and vitreous humor. - Stickler syndrome → congenital hearing loss.
Type III	- Found in blood vessels; early wound healing (pink in color); ↓ tensile strength compared to type I. - Ehlers-Danlos (vascular type) → hyperextensible skin/joints; easy bruising; aortic dissection/regurgitation; mitral valve prolapse (myxomatous degeneration); circle of Willis berry (saccular) aneurysms.
Type IV	- Found in basement membranes of the kidney + alveoli; also found in the lens, cornea, and inner ear. - Alport (XR; *mutation in* type IV collagen) → eye/ear problems in male with hematuria. - Goodpasture (*Abs against* type IV collagen; aka anti-GBM antibodies) → male 20s-40s hemoptysis + hematuria.

Shock types			
	Cardiac output (CO)	Systemic vascular resistance (SVR)	Pulmonary capillary wedge pressure (PCWP)
Cardiogenic	↓	↑	↑
Hypovolemic	↓	↑	↓
Septic	↑ (early) / ↓ (late)	↓	↓
Anaphylactic	↑ (early) / ↓ (late)	↓	↓
Neurogenic	↓	↓	↓
Obstructive	↓	↑	↓
- Septic, anaphylactic, and neurogenic are all under the envelope of distributive shock. - ↑ PCWP for cardiogenic is one of the highest yield path points on USMLE. - For deeper explanations, go to my HY Arrows PDF.			

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